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Complete List of Publications

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Summary

14 peer-reviewed journal papers | 11 conference proceedings | 1 book | 2 book chapters | 3 dissertations.

Bibliometrics. h-index **7** on both indices. Web of Science (September 2026): 122 citations across 25 indexed publications, 17 of them in the Core Collection. Google Scholar: 184 citations.

Entries are reverse chronological within each section. Numbering runs continuously across the whole list. Author name in bold throughout.

Journal Papers

1. V. Joseph, B.K. Sarkar, A. Chatterjee, **S. Chaudhuri**, “Fast nonsingular terminal sliding mode controller design for Francis turbine with electrohydraulic IGV: an experimental investigation,” *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, January 2026. DOI: [10.1177/09596518251399972](#)
2. Y. Sheikh Kilo, R. Izadi-Zamanabadi, H. Ramezani, P. Gullo, **S. Chaudhuri**, “Cost-effective optimization for charging/discharging cycles of thermal energy ice storages in transcritical R744 supermarket refrigeration systems,” *Energy Conversion and Management*, vol. 334, 119790, June 2025. ISSN 0196-8904. DOI: [10.1016/j.enconman.2025.119790](#)
3. S. Paul, S. Das, M. Banerjee, **S. Chaudhuri**, B. Das, “The ATP-dependent DEAD-box RNA helicase Dbp2 regulates the glucose/nitrogen stress response in baker’s yeast by modulating reversible nuclear retention and decay of *SKS1* mRNA,” *Genetics*, vol. 229, no. 3, iyae221, March 2025. DOI: [10.1093/genetics/iyae221](#)
4. H. Ramezani, **S. Chaudhuri**, J. Jouffroy, A. Baurichter, S.M. Hansen, “On roll stabilisation using a canting keel,” *IEEE Transactions on Control Systems Technology*, vol. 33, no. 2, pp. 539–553, March 2025 (early access November 2024). ISSN 1558-0865. DOI: [10.1109/TCST.2024.3494237](#)
5. N. Taghva, **S. Chaudhuri**, “How language-specific and cross-linguistic factors affect speech rhythm,” *Estudios de Fonética Experimental*, vol. 33, pp. 169–190, September 2024. DOI: [10.1344/EFE-2024-33-169-190](#)
6. A. Sarkar, S. Dasmahapatra, **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, “A novel order-separated generalized feedforward design for motion control in energy-efficient electrohydraulic systems with proportional and integral feedback,” *ISA Transactions*, vol. 146, pp. 336–351, January 2024. ISSN 0019-0578. DOI: [10.1016/j.isatra.2024.01.012](#)
7. I. De, S. Sarkar, **S. Chaudhuri**, N. Mondal, N. Kumar, “Effect of dynamic swivelling torque and eccentricity on the design of compensator cylinders for a variable displacement axial piston pump: modelling and simulation,” *Jordan Journal of Mechanical and Industrial Engineering*, vol. 17, no. 2, pp. 255–268, June 2023. ISSN 1995-6665. DOI: [10.59038/jjmie/170209](#)
8. A. Sarkar, K. Maji, **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, “Actuation of an electrohydraulic manipulator with a novel feedforward compensation scheme and PID feedback in servo-proportional valves,” *Control Engineering Practice*, vol. 135, 105490, March 2023. ISSN 0967-0661. DOI: [10.1016/j.conengprac.2023.105490](#)
9. S. Sarkar, R. Mandal, N. Mondal, **S. Chaudhuri**, T. Mandal, G. Majumdar, “Modelling and prediction of micro-hardness of electroless Ni-P coatings using response surface methodology and fuzzy logic,” *Jordan Journal of Mechanical and Industrial Engineering*, vol. 16, no. 5, pp. 729–742, December 2022. ISSN 1995-6665.
10. P. Datta, A. Chakravarty, R. Saha, **S. Chaudhuri**, K. Ghosh, A. Mukhopadhyay, S. Sen, A. Dutta, P. Goyal,

“Experimental investigation on the effect of initial pressure conditions during steam-water direct contact condensation in a horizontal pipe geometry,” *International Communications in Heat and Mass Transfer*, vol. 121, 105082, February 2021. ISSN 0735-1933.

DOI: [10.1016/j.icheatmasstransfer.2020.105082](https://doi.org/10.1016/j.icheatmasstransfer.2020.105082)

11. **S. Chaudhuri**, R. Saha, A. Chatterjee, S. Mookherjee, D. Sanyal, “Adaptive neural-bias-sliding mode control of rugged electrohydraulic system motion by recurrent Hermite neural network,” *Control Engineering Practice*, vol. 103, 104588, October 2020. ISSN 0967-0661.
DOI: [10.1016/j.conengprac.2020.104588](https://doi.org/10.1016/j.conengprac.2020.104588)
12. **S. Chaudhuri**, R. Saha, A. Chatterjee, S. Mookherjee, D. Sanyal, “Development of a motion sensing system based on visual servoing of an eye-in-hand electrohydraulic parallel manipulator,” *IEEE Sensors Journal*, vol. 20, no. 14, pp. 8108–8116, July 2020. ISSN 1558-1748.
DOI: [10.1109/JSEN.2020.2979490](https://doi.org/10.1109/JSEN.2020.2979490)
13. S. Sarkar, A. Mukherjee, S. Das, B. Ghosh, **S. Chaudhuri**, D. Bhattacharya, A. Sarbajna, R. Gachhui, “Nitrogen deprivation elicits dimorphism, capsule biosynthesis and autophagy in *Papiliotrema laurentii* strain RY1,” *Micron*, vol. 124, 102708, September 2019. ISSN 0968-4328.
DOI: [10.1016/j.micron.2019.102708](https://doi.org/10.1016/j.micron.2019.102708)
14. S. Das, S. Biswas, **S. Chaudhuri**, A. Bhattacharyya, B. Das, “A nuclear zip code in *SKS1* mRNA promotes its slow export, nuclear retention, and degradation by the nuclear exosome/DRN in *Saccharomyces cerevisiae*,” *Journal of Molecular Biology*, vol. 431, no. 19, pp. 3626–3646, September 2019. ISSN 0022-2836.
DOI: [10.1016/j.jmb.2019.07.005](https://doi.org/10.1016/j.jmb.2019.07.005)

Conference Proceedings

15. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, “Ensuring safe roll dynamics in marine vessels based on a safety-enhanced control framework,” *International Conference on Electrical, Computer and Energy Technologies (ICECET)*, 2026. *Presented*.
16. Z. Gniewosz, **S. Chaudhuri**, H. Ramezani, “Torque vectoring-based active yaw control for enhanced vehicle handling and stability using a digital twin framework,” *International Conference on Mechanical Engineering (INCOM)*, Kolkata, India, January 2026. *Presented*.
17. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, “Remarks on augmenting stability and safety in nominal controllers: a practical perspective,” *European Control Conference (ECC)*, Thessaloniki, Greece, pp. 2121–2128, 2025.
DOI: [10.23919/ECC65951.2025.11187095](https://doi.org/10.23919/ECC65951.2025.11187095)
18. H. Ramezani, **S. Chaudhuri**, J. Jouffroy, A. Baurichter, S.M. Hansen, “On roll stabilisation using a canting keel,” *62nd IEEE Conference on Decision and Control (CDC)*, Singapore, 13–15 December 2023.
DOI: [10.1109/CDC49753.2023.10384049](https://doi.org/10.1109/CDC49753.2023.10384049)
19. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, “Investigation of marine vessels equipped with an airkeel subject to wave-induced disturbances,” *IEEE Conference on Control Technology and Applications (CCTA)*, Bridgetown, Barbados, pp. 484–489, 16–18 August 2023.
DOI: [10.1109/CCTA54093.2023.10252537](https://doi.org/10.1109/CCTA54093.2023.10252537)
20. **S. Chaudhuri**, H. Ramezani, J. Jouffroy, “Modelling and control of a canting-keel based ship roll stabilization system for crane operations,” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Prague, Czech Republic, pp. 1904–1909, 9–12 October 2022.
DOI: [10.1109/SMC53654.2022.9945570](https://doi.org/10.1109/SMC53654.2022.9945570)
21. R. Bhattacharjee, S. Kundu, **S. Chaudhuri**, “Evaluation of workspace and coupled motions of an electrohydraulic parallel manipulator,” *Fourth International Conference on Electrical, Computer and Communication Technologies (ICECCT)*, Erode, India, pp. 1–8, 15–17 September 2021.
DOI: [10.1109/ICECCT52121.2021.9616672](https://doi.org/10.1109/ICECCT52121.2021.9616672)
22. S. Kundu, R. Bhattacharjee, **S. Chaudhuri**, “Evaluation of fuzzy-logic based position control strategies for an electrohydraulic actuation system,” *International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, Bhilai, India, pp. 1–7, 19–20 February 2021.
DOI: [10.1109/ICAECT49130.2021.9392479](https://doi.org/10.1109/ICAECT49130.2021.9392479)
23. **S. Chaudhuri**, R. Saha, S. Mookherjee, D. Sanyal, A. Chatterjee, “Visual sensing based adaptive sliding mode control of position tracking in electrohydraulic systems,” *Second International Conference on Control, Instrumentation, Energy & Communication (CIEC)*, Kolkata, India, pp. 25–29, 28–30 January 2016.
DOI: [10.1109/CIEC.2016.7513762](https://doi.org/10.1109/CIEC.2016.7513762)

24. **S. Chaudhuri**, S. Mookherjee, D. Sanyal, “Adaptive force tracking in electrohydraulic system with first-order sliding mode control,” *IEEE First International Conference on Control, Measurement and Instrumentation (CMI)*, Kolkata, India, pp. 300–304, 8–10 January 2016.
DOI: [10.1109/CMI.2016.7413759](https://doi.org/10.1109/CMI.2016.7413759)
25. S. Dasmahapatra, **S. Chaudhuri**, P. Mandal, S. Mookherjee, R. Saha, “Fuzzy-PI control of motion tracking by an electrohydraulic system with multiple nonlinearities,” *Michael Faraday IET International Summit*, Kolkata, India, 12–13 September 2015.
DOI: [10.1049/cp.2015.1612](https://doi.org/10.1049/cp.2015.1612)

Book

26. **S. Chaudhuri**, *Pressurised Water Nuclear Reactor: Dynamics, Modelling and Simulation*, LAP Lambert Academic Publishing (OmniScriptum Publishing Group), 164 pp., 2019. ISBN 978-620-0-09362-2.

Book Chapters

27. R. Bhattacharjee, **S. Chaudhuri**, A. Ganguly, “Robust control of pulsatile ventricular assist devices for patients with advanced heart failure,” in *Proceedings of the International Conference on Frontiers in Computing and Systems (COMSYS 2022)*, Lecture Notes in Networks and Systems, Springer, Singapore, pp. 331–341, 2023. ISBN 978-981-99-2679-4.
DOI: [10.1007/978-981-99-2680-0_29](https://doi.org/10.1007/978-981-99-2680-0_29)
28. **S. Chaudhuri**, S. Dasmahapatra, A. Chatterjee, R. Saha, S. Mookherjee, D. Sanyal, “Adaptive fuzzy-sliding mode control with fixed bias compensator for an electrohydraulic actuation system with hard nonlinearities,” in *Fluid Mechanics and Fluid Power: Contemporary Research*, Lecture Notes in Mechanical Engineering, Springer, New Delhi, pp. 1223–1232, 2017. ISBN 978-81-322-2743-4.
DOI: [10.1007/978-81-322-2743-4_116](https://doi.org/10.1007/978-81-322-2743-4_116)

Dissertations

29. **S. Chaudhuri**, *Developing Electrohydraulic System Solutions using Adaptive-Neuro-Sliding Mode Control and Vision Sensing*, doctoral dissertation, Department of Electrical Engineering, Jadavpur University, Kolkata, India, 2021. Supervisors: Prof. Amitava Chatterjee and Prof. Saikat Mookherjee.
30. **S. Chaudhuri**, *Compact Reactor Modelling*, postgraduate dissertation, M.E. in Nuclear Engineering, Jadavpur University, carried out at the Bhabha Atomic Research Centre (BARC), Mumbai, India, 2013. Advisor: Dr. Siddhartha Mukhopadhyay.
31. **S. Chaudhuri**, *Study and Fabrication of a Battery Low Indicator*, undergraduate dissertation, Seacom Engineering College, West Bengal University of Technology, India, 2011. Advisor: Mr. Sarbojit Mukherjee.